Adigopula Rohith Ganesh

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PROFESSIONAL SUMMARY

Systems Engineer at Tata Consultancy Services specializing in Generative AI, Agentic AI, and secure backend development using Python, FastAPI, LangGraph, LangChain, and MCP. Experienced in patient-specific RAG, LLM evaluation, vector retrieval, stateful orchestration, and AWS integration. Engineered AI solutions evaluated through retrieval accuracy, response grounding, access-control tests, and API efficiency.

SKILLS

- Programming Languages: Python, Java, SQL
- Generative and Agentic AI: LangGraph, LangChain, MCP, RAG, Prompt Engineering, RAGAS, LangSmith
- Backend and APIs: FastAPI, Django, Spring Boot, Spring MVC, REST APIs, Spring Data JPA, Hibernate, JWT, RBAC
- Databases and Vector Stores: MongoDB Atlas Vector Search, ChromaDB, MySQL, PostgreSQL, MongoDB
- Cloud and Development Tools: AWS, Docker, Git, GitHub, Postman
- Software Engineering: System Design, Debugging, Code Review, Algorithms, Data Structures, Feature Implementation

EXPERIENCE

Systems Engineer | **Tata Consultancy Services (TCS)** **Dec 2025 – Present**

- Engineered Python and FastAPI services for healthcare AI workflows spanning document retrieval, relevance validation, and grounded-answer generation across 3 core processing stages.
- Implemented 2 complementary access-control mechanisms—JWT authentication and role-based authorization—alongside patient-scoped retrieval and conditional LangGraph routing to restrict unauthorized data access.
- Integrated MCP-compatible tools and enterprise APIs with controlled execution, input validation, and retrieval-grounding safeguards for healthcare-focused AI workflows.

Intern | **Cognizant** **Jan 2025 – Jul 2025**

- Implemented Java backend components using Spring Boot, Spring Data JPA, Hibernate, and MySQL, integrating 3 application layers—REST APIs, business services, and persistence— for healthcare workflows.
- Supported healthcare application development through database-backed CRUD operations, API integration, debugging, and maintainable service-layer implementations.

PROJECTS

JobAlign – Intelligent Job Search Platform **Jun 2026 – Jul 2026**

- Built an LLM-powered job platform for resume analysis, skill-gap detection, and semantic job matching; reported 80% Precision@5 across 100 manually labeled job recommendations.
- Reduced Tavily search calls from 3 to 2 per request (33.3% reduction) and limited retrieved results from 10 to 6 (40% fewer results processed), while supporting AI-generated cover letters and recruiter messages.
- Tech Stack: Python, FastAPI, LangChain, Groq, Tavily, ChromaDB, Hugging Face, React.

PatientScope – Patient-Specific Medical RAG **Jul 2026 – Aug 2026**

- Architected a patient-specific medical RAG assistant supporting 7 clinical document categories; reported 88% grounded-answer correctness across 50 synthetic evaluation questions using RAGAS faithfulness and reference-answer review.
- Achieved a reported 100% pass rate (50/50 tests) for patient-isolation scenarios involving cross-patient retrieval, unauthorized document access, and patient-ID manipulation; integrated Presidio PHI masking, JWT authentication, and RBAC.
- Tech Stack: Python, FastAPI, LangGraph, LangChain, MongoDB Atlas, Hugging Face, Presidio, Groq, JWT.

CloudLens MCP – AWS Account Intelligence **Aug 2026 – Sep 2026**

- Developed an AWS account intelligence architecture for cost auditing, security inspection, and operational monitoring using MCP-based natural-language tooling and serverless infrastructure.
- Designed read-only access across 5 AWS service domains—EC2, S3, IAM, CloudWatch, and Cost Explorer— using permission-scoped tool execution, with a React dashboard for findings, resource visibility, and cost insights.
- Tech Stack: Python, MCP, AWS Lambda, API Gateway, IAM, CloudWatch, Cost Explorer, React.

EDUCATION

K L University **2021 – 2025**

Bachelor of Technology in Computer Science and Engineering **CGPA: 9.47/10**

CERTIFICATIONS

- AWS Certified AI Practitioner
- Claude Certified Architect – Foundations